

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations

December 2025

EC2105-1 – NETWORK THEORY AND CONTROL SYSTEMS

(For ECE, ACT, VDT)

Duration: 3 Hours

Max. Marks: 100

Note:

Usage of Graph sheet and Semilog Sheet is permitted

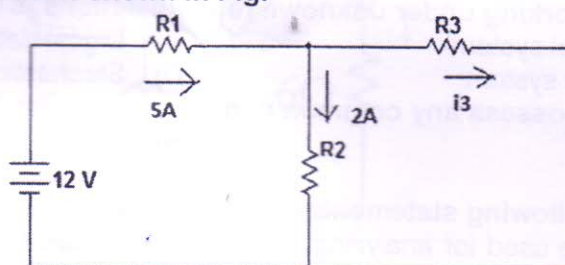
Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Which law plays a significant role in the mesh analysis of the network?
 A) KVL
 B) KCL
 C) Law of superposition
 D) Law of Reciprocity
2. Find the current i_3 in the circuit shown in Fig.



- A) 5A
 B) 2A
 C) 3A
 D) 1A
3. A star connection has each element of resistance $\frac{R}{2}$. The equivalent resistance in delta connection will be
 A) $\frac{R}{6}$
 B) $\frac{3R}{2}$
 C) $2R$
 D) $4R$
4. Potential difference in electrical terminology is known as?
 A) Voltage
 B) Resistance
 C) Current
 D) Conductance
5. Applying Norton's theorem to a linear two-terminal network yields an equivalent circuit consisting of
 A) A single equivalent current source and an impedance in series
 B) A single equivalent current source and an impedance in parallel
 C) A single equivalent impedance
 D) A single equivalent voltage source and an impedance in parallel
6. Thevenin's voltage is equal to the _____ voltage across the _____ terminals.
 A) open circuit, output
 B) open circuit, input
 C) short circuit, input
 D) short circuit, output

7. Thevenin's equivalent impedance for a circuit is found to be $Z_s = 4 - 3j$. The power transferred will be maximum when the load impedance is equal to
 A) $4 - 3j$ B) $3 - 4j$
 C) $3 + 4j$ D) $4 + 3j$
8. While applying superposition theorem, dependent sources are
 A) Replaced with short circuit B) Replaced with open circuit
 C) Not altered D) Dependent current sources are replaced with an open circuit and dependent voltage sources are replaced with a short circuit
9. In two-port networks the parameter h_{22} is called
 A) Open circuit output admittance B) Open circuit reverse voltage gain
 C) Short circuit input impedance D) Short circuit current gain
10. The unit in which the Z-parameters are expressed is
 A) Mho B) Ohm
 C) No Unit D) Farad
11. In determining short circuit admittance parameters, among V_1, V_2, I_1, I_2 , which of the following are dependent variables?
 A) V_1 and V_2 B) I_1 and V_2
 C) I_1 and I_2 D) V_1 and I_2
12. Which of the following expression is true in case of open circuit impedance parameters?
 A) $V_1 = Z_{11} I_1 + Z_{12} V_2$ B) $V_1 = Z_{11} I_1 + Z_{12} I_2$
 C) $V_1 = Z_{11} I_2 + Z_{12} V_2$ D) $V_2 = Z_{11} V_1 + Z_{12} I_2$
13. A control system working under unknown random actions is called
 A) Computer control system B) Digital data system
 C) Adaptive control system D) Stochastic control system
14. Loops that do not possess any common node are said to be _____ loops.
 A) Forward gain B) Touching
 C) Non touching D) Feedback gain
15. Which one of the following statements is not correct?
 A) Root loci can be used for analyzing stability and transient performance B) Shape of the root locus gives idea of type of controller needed to meet design specification
 C) Root loci provide insight into system stability and performance D) Root locus can be used to handle more than one variable at a time
16. Root locus is used to calculate
 A) Absolute stability B) Relative stability
 C) Conditional stability D) Marginal stability
17. Initial slope of +40db/decade in the magnitude Bode plot indicates the presence of
 A) Single pole at origin B) Two poles at origin
 C) Single zero at origin D) Two zeros at origin
18. Gain margin of a stable first or second order system is
 A) 0 B) 10
 C) 1 D) ∞ (Infinity)
19. The frequency at which the magnitude plot crosses 0 dB line is called
 A) Gain crossover frequency B) Phase margin frequency
 C) Gain margin frequency D) Phase crossover frequency
20. In the Bode plot, the frequency where the two asymptotic lines meet is called
 A) Linear frequency B) Gain cross over frequency
 C) Corner frequency D) Phase cross over frequency

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

Marks BT* CO* PO*

1. a) For the circuit shown in Fig. Q1 (a), find i_1 using source transformation.

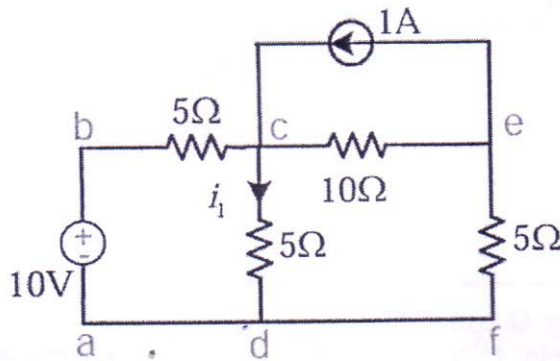


Fig. Q1 (a)

8 L*3 1 1

- b) Using superposition theorem, find the value of I for the circuit shown in Fig. Q 1(b).

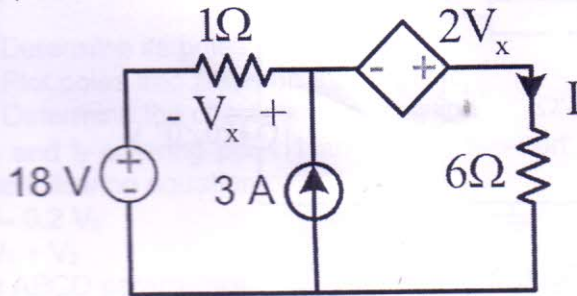


Fig. Q 1(b)

8 L3 2 1

2. a) Using mesh current analysis, find the current through 10 Ω resistor for the circuit shown in Fig. Q2 (a).

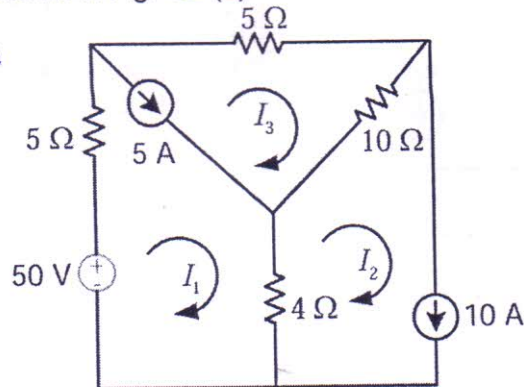


Fig. Q2 (a)

8 L3 1 1

- b) What should be the value of the pure resistance to be connected across the terminals A – B in the network shown in Fig. Q 2(b) so that the maximum power is transferred to the load? What is the maximum power?

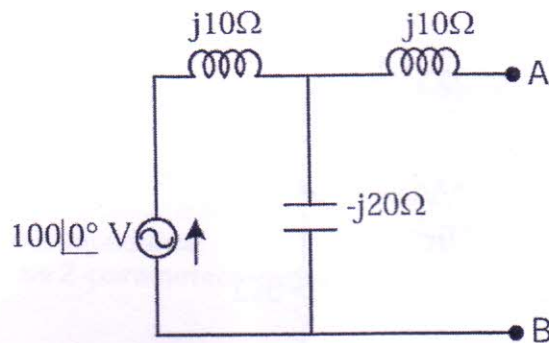


Fig. Q 2(b)

8 L3 2 1

3. a) Consider the circuit shown in Fig. Q 3(a). Find the value of I using node voltage analysis.

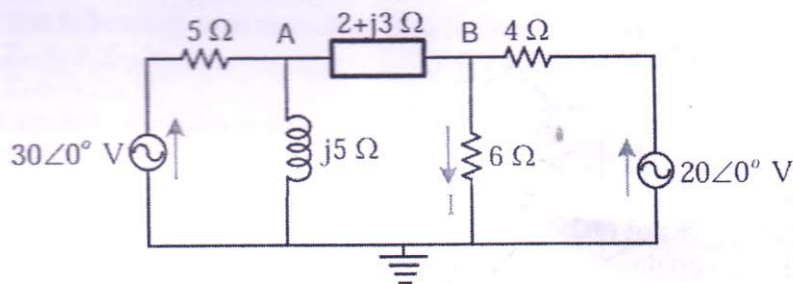


Fig. Q 3(a)

8 L3 1 1

- b) Calculate the current I_L flowing through the load R_L for the circuit shown in Fig. Q 3(b) using Norton's theorem.

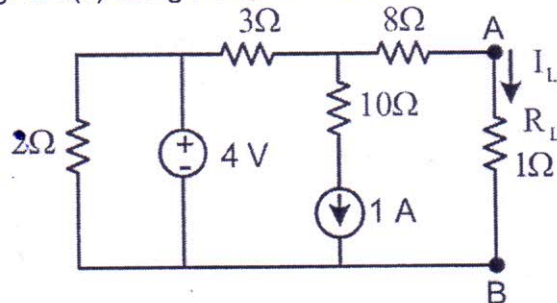


Fig. Q 3(b)

8 L3 2 1

Unit – II

4. a) Find the transmission parameters for the circuit shown in Fig. Q 4(a).

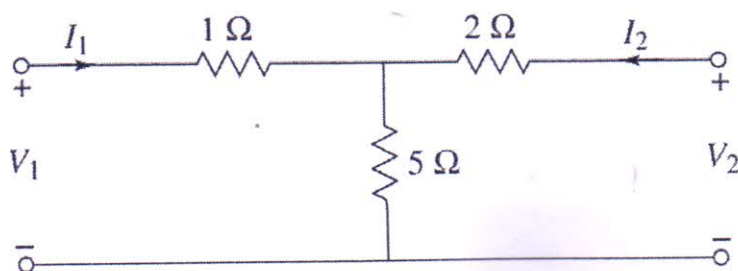


Fig. Q 4(a)

8 L3 3 1

- b) Find the range of values of 'K' so that the system with the following characteristic equation will be stable:
- $$s(s^2 + s + 1)(s + 4) + K = 0$$

8 L3 4 1

5. a) Consider the Signal Flow Graph shown in Fig. Q 5(a). Find the overall transfer function using Mason's Gain Formula.

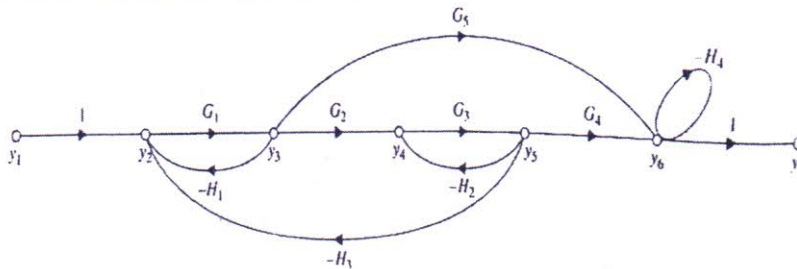


Fig. Q 5(a)

8 L3 4 1

- b) Consider the transfer function of a system given by

$$T(S) = \frac{K(s+6)}{s(s+2)(s+5)(s^2+7s+2)}$$

- Determine its poles and zeros
- Plot poles and zeros on the S- plane
- Determine the characteristic equation

8 L3 4 1

6. a) Currents I_1 and I_2 entering ports 1 and 2 of a two-port network are given by the following equations:

$$I_1 = 0.5 V_1 - 0.2 V_2$$

$$I_2 = -0.2 V_1 + V_2$$

Find Y and ABCD parameters.

6 L3 3 1

- b) Sketch the complete root locus of a system with

$$G(S)H(S) = \frac{K}{s(s+1)(s+2)(s+3)}$$

Comment on its stability.

10 L3 4 1

Unit – III

7. a) Consider the magnitude plot of system shown in Fig. Q 7(a). Determine its transfer function.

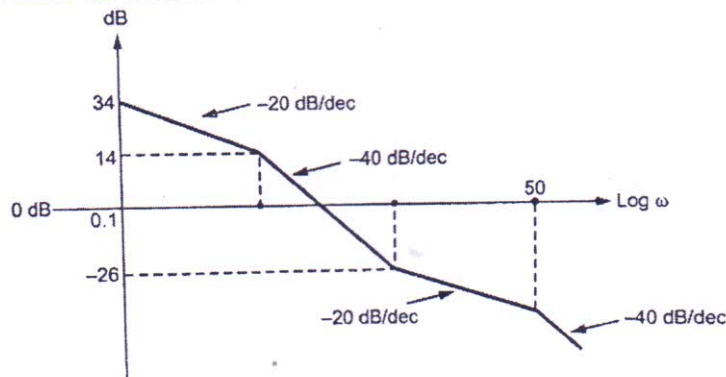


Fig. Q 7(a)

6 L3 5 1

- b) Consider a unity feedback system with $G(S) = \frac{800(s+2)}{s^2(s+10)(s+40)}$

Sketch the Bode plot. Comment on its stability.

10 L3 5 1

8. a) Obtain the transfer function of a unity feedback system whose magnitude plot is as given in Fig. Q 8(a).

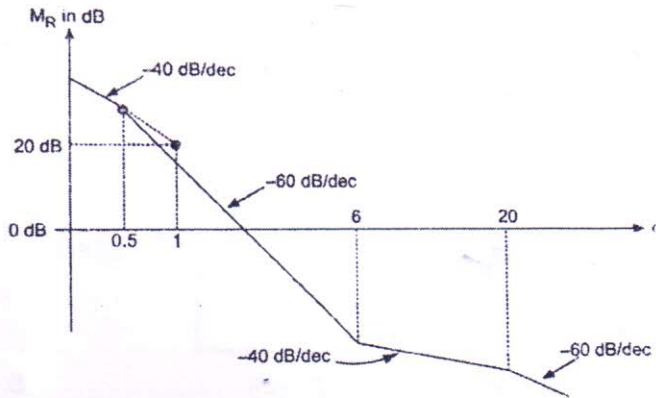


Fig. Q 8(a)

- b) For a unity feedback system has $G(S) = \frac{80}{s(s+2)(s+20)}$

Sketch the Bode plot. Comment on its stability.

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.

6 L3 5 1

10 L3 5 1